① Req DNS for IP addr of "abc.com"

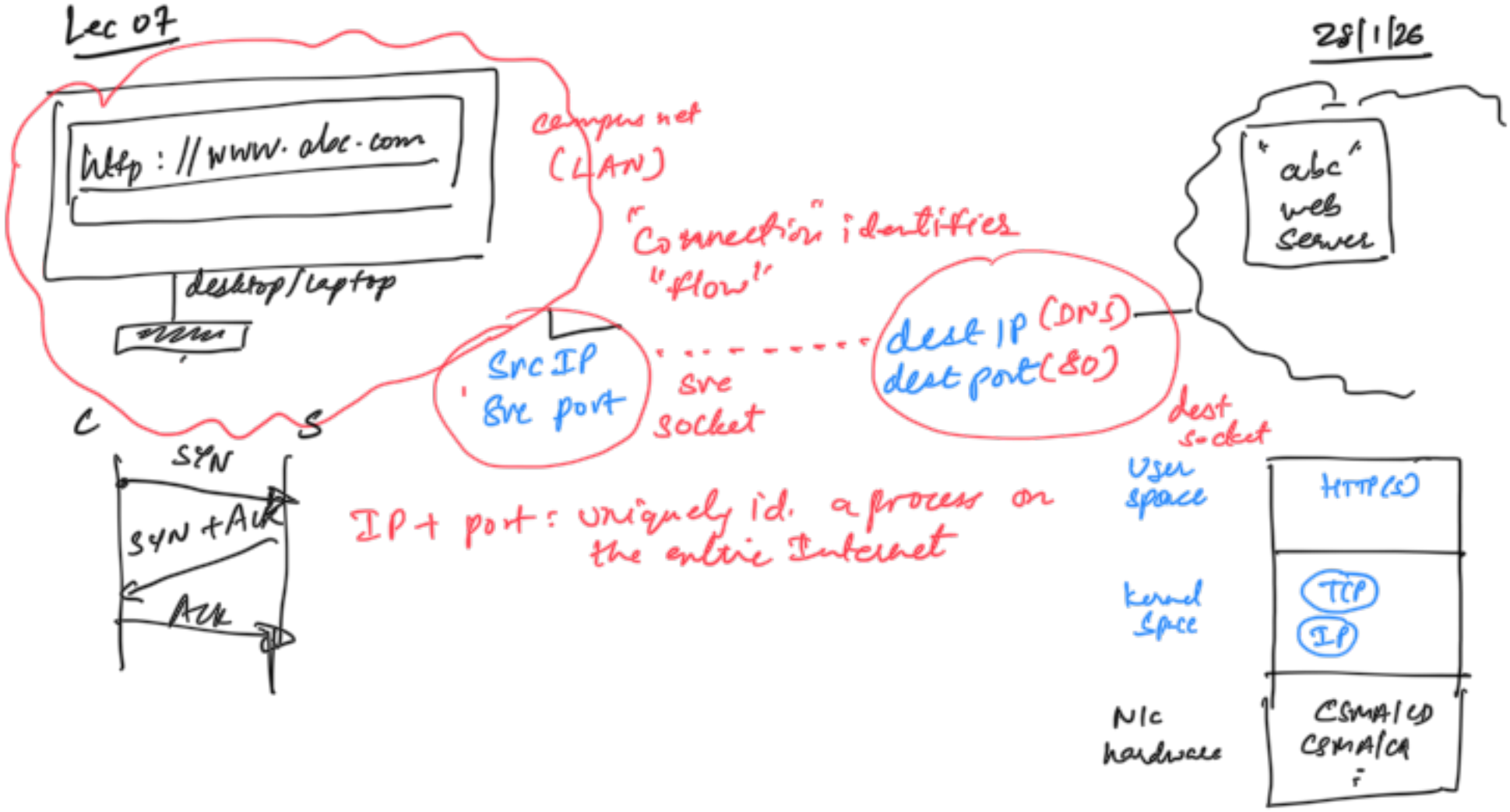
→ ② Open a TCP conn" with "abc.com" IP

- DNS IP
- Default gateway
- IP addr of mlc

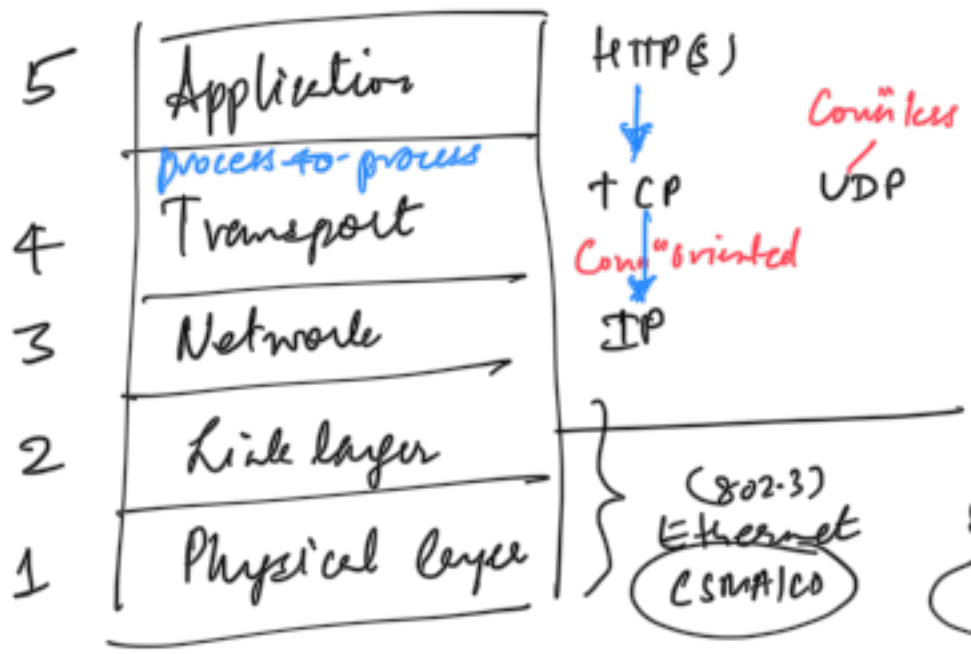
HTTP GET/POST/PUT/DELETE ....

Lec 07

28/1/26



* TCP/IP protocol stack / Networking Stack ... adopted by Internet




- reliable
- packet ordering

- Conn" oriented
- Conn"less (best-effort)
  - ↳ Acked
  - ↳ unacked.

- Addresses
  ① Physical addr (MAC addr [48-bit addr ... NIC]
  unique id every machine on a local network
  ② Logical addr (eg. IP addr) (IPv4: 32bit IPv6: 128b.)
  Unique id every m/c global (Internet)

③ Port address (16-bit)
Unique id for every process in a machine

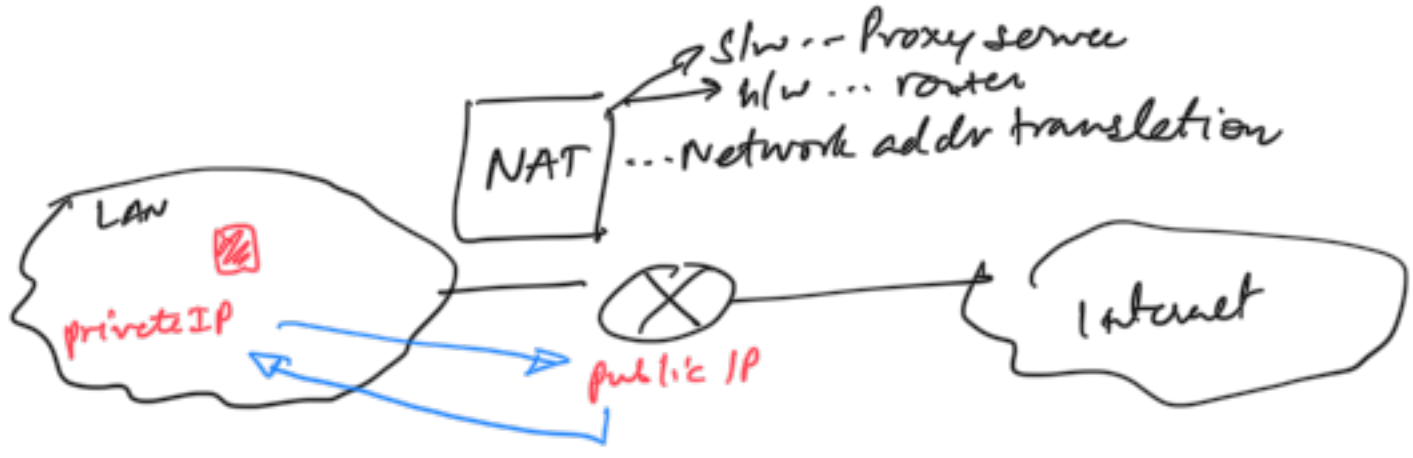
HTTP: 80
HTTPS: 443

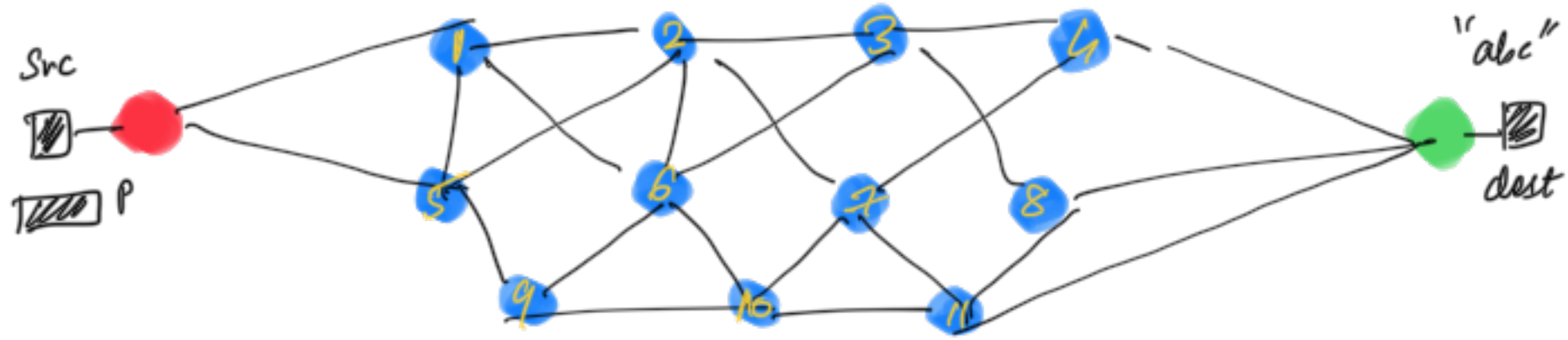
- < 1024 .. std port addr [rsvd]
- above 1024 .. user-defined paddr

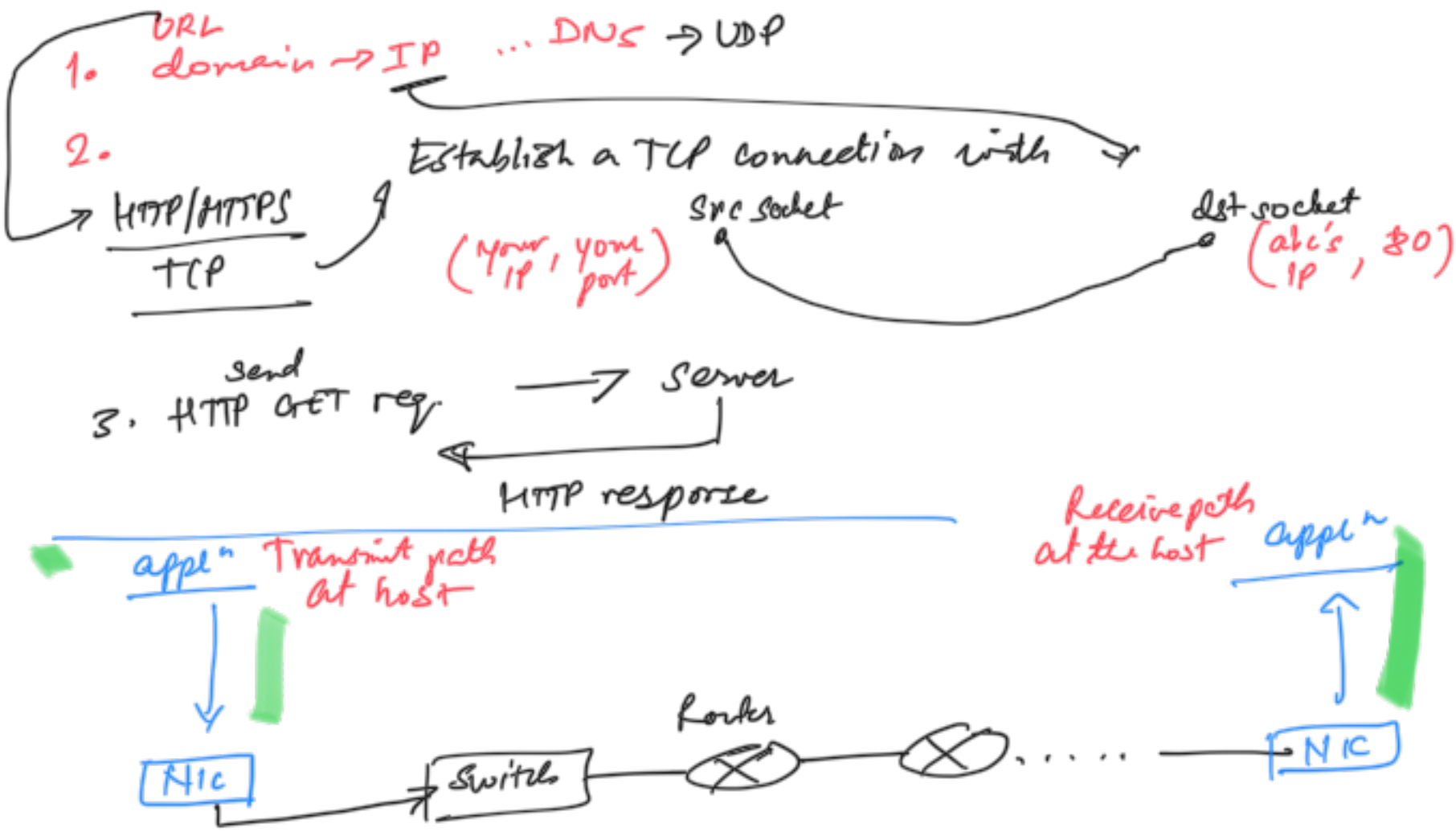
Public IP : Unique globally
↳ Private IP : (private) unique within local n/w. → 10.x.x.x
172.16.x.x
192.168.x.x

IPv4: $2^{32} = 4 \times 10^9$ addr

↳ limited IP range [scaling #m/c.]




Src
P
1
2
3
4
5
6
7
8
9
10
11
"abc"
dest


URL
1. domain → IP ... DNS → UDP
2.
HTTP/HTTPS
TCP
Establish a TCP connection with
(Your IP, your port)
Src socket
Dst socket
(abc's IP, 80)
send
3. HTTP GET req.
server
HTTP response
appln
Transmit path at host
Receive path at the host
appln
Router
NIC
Switch
NIC